

DATASHEET / LIST

Line / Valve List Auto Generator from P&ID & PMC

Builds a LINE LIST automatically by decoding KKS tags from P&ID (PDF/DXF). Includes From/To derivation and valve/symbol recognition.

Key Purpose

Why this program

1. Build the LINE LIST straight from the P&ID

- Decodes KKS line tags from a PDF P&ID and lays out every line automatically

→ **Effect:** A full line list that once took days of manual transcription is produced in minutes with no typing errors

2. Derive From/To and recognise valves

- Traces connectivity for From/To endpoints and identifies valves/symbols to also emit a VALV LIST

→ **Effect:** Connectivity and valve data are filled in consistently, so the list is review-ready instead of half-empty

3. Enrich each line with class, material and operating P/T

- Optional PMC legend maps BB1 codes to class/material; Performance Summary supplies operating pressure/temperature

→ **Effect:** Design conditions are pulled from source documents automatically, keeping every line traceable and spec-complete

Required inputs : P&ID (PDF) + optional PMC Legend (PDF/XLSM) + optional Performance Summary (PDF)

Main Window – Item Guide

The screenshot shows the main window of the 'LINE / VALVE LIST Generator - from P&ID' application. The window has a title bar and a menu bar. Below the menu bar, there are four input sections, each with a text field and a 'Browse...' button. The first section is for the 'P&ID File (PDF / DXF):' with a text field (1) and a 'Browse...' button (2). The second section is for the 'PMC Legend (PDF / XLSM — BB1→Class/Material, optional):' with a text field (3) and a 'Browse...' button. The third section is for the 'Performance Summary (PDF — Operating P/T, optional):' with a text field (4) and a 'Browse...' button. The fourth section is for the 'Output XLSX (default: same folder as P&ID):' with a text field (5) containing the path 'C:\Users\Wacepl\OneDrive\JJ\DATA\THON\DATA_SHEET\LINE_LIST\WO' and a 'Browse...' button (6). Below these sections are four buttons: 'Generate LINE LIST' (7), 'Open Output File' (8), 'Generate VALV LIST' (9), and 'Open Output File' (10). At the bottom of the window is a large empty rectangular area (11) for a log or progress display.

1. P&ID File (PDF/DXF) — main input
2. Browse for P&ID file
3. PMC Legend (BB1 to Class/Material, optional)
4. Performance Summary (Operating P/T, optional)
5. Output XLSX path (default: P&ID folder)
6. Browse for output XLSX path
7. Generate LINE LIST
8. Open LINE LIST output file
9. Generate VALV LIST
10. Open VALV LIST output file
11. Log — progress & results

Overview

What it does

1. Builds a LINE LIST automatically by decoding KKS tags from P&ID (PDF/DXF)
2. Includes From/To derivation and valve/symbol recognition

Category: Datasheet / List

Folder: DATA_SHEET/LINE_LIST

Main scripts: linelist_gui.py

Scripts / Status: 15 scripts · Operational

Tags: P&ID · KKS · LineList · PDF

Output Samples

[illegible]

Line list Auto Gen Output sample

[illegible]

Valve list Auto Gen Output sample

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Datasheet / List.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: `linelist_gui.py`.
- Output samples and the program window above reflect the current build.

Thank You

Line / Valve List Auto Generator from P&ID & PMC · Plant Engineering Total Service System